
Final Lab
Final Project
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ECEN 2611: Instrumentation and Computation Lab 1

Lab Report

Dylan Okular

Aedan Shamblin

Muhammad Al Faqeeh Adebayo

Instructor

Prof. Teresa Ren

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Objective:

Performing circuit analysis to get desired theoretical results and then comparing the results with both measured experimental and simulated data.

Introduction:

This lab began with a brief quiz over Part B of the final project, we had to find the Thevenin equivalent of a circuit as well as its Norton current as well. From here we began this lab by being informed that we were to only rely on our own knowledge and our partners. We were not permitted to ask for any help from instructors or UAs.

Details:

The materials required for this experiment consists of:

- 1x Digital Multimeter(DMM)
- 1x DC Power Supply
- 1x RC Box
- 2x $1k\Omega$ Resistor
- 1x $2.2k\Omega$ Resistor
- 1x 510Ω Resistor
- 1x 10Ω Resistor
- 1x 33Ω Resistor
- 1x 51Ω Resistor
- 1x 47Ω Resistor
- 1x 22Ω Resistor
- Banana-Alligator Wires
- Jumper Wires
- Banana Wires
- 1x MyDigital Protoboard

To begin this lab we first must calculate the current flowing through each resistor in the circuit depicted in Figure 1. To do this you will need to use mesh and nodal analysis to determine each value, once you determine your current values you will need to record them in Column 2 of Table 1 (your work should appear similar to Figure 3.) Following these calculations you will need to construct the circuit in Figure 1 on your MyDigital Protoboard to measure the currents across each resistor using the DMM. To do this you will first need to attach your resistors to the board, do this by attaching your 22Ω resistor in series with both your 47Ω and 51Ω resistors. Then having your 33Ω resistor be in series with both the other terminal of the 51Ω resistor and 10Ω resistor. Then attach a wire connecting the bottom terminals of both the 47Ω and 10Ω , next attach the ground from the DC Power Supply in the same column as one of the terminals of the wire you just connected.

Now attach your 1 VDC and 2VDC channels from the DC power supply to the 22Ω resistor and 33Ω resistor respectively (see Figure 4), after enabling the flow of current on the DC power supply use the DMM as an ammeter to determine the current across each resistor and record your measurements in column 3 of Table 1. Now moving to Part B, we begin by finding the Thevenin and Norton equivalent models of the circuit shown in Figure 2, record your calculations in column 2 of Table 2. Next, we will construct this circuit in PSpice to determine the simulated values. To do this we will first place our resistors and load, attaching our $1k\Omega$ resistor in series with both or $2.2k\Omega$ and 510Ω resistors, then attaching a $1k\Omega$ in series with our 510Ω and then our load in parallel with our $1k\Omega$ resistor. Then attach a 10VDC source with the positive terminal attached to the $1k\Omega$ and the negative terminal attached to the bottom terminal of the $2.2k\Omega$ resistor, your final circuit should appear similar to Figure 5.

Use this circuit to record the simulated values in column 3 of Table 2, now following the same steps for laying out the simulation create this circuit on your MyDigital Protoboard. Our load will be our RC Box, and our DC Power Supply will be our VDC source, ensure everything is grounded and then complete Table 2 by filling in column 4. Now, to begin to prepare our circuit schematic for board layout, we start this by creating a new schematic under our .dsn file in the project hierarchy. This will create a new folder on the hierarchy, and we can then create a new page to copy our initial circuit into, both files should be named "PCBSCH_yourname".

On this new page we want to paste our circuit and alter it by adding a connector using the place part menu in PSpice. Now we need to press "Ctrl+A" and right click and select "Edit Properties" to attach footprints to our components. Once in our menu we want to scroll until we see the column labeled "PCB Footprint" and we want to change the footprint of R1, R2, R3, R4, and R5 to say "res400" and change the footprint of J1 to "jumper2". From here we can go to file and save our schematic and then make the folder we just made the root of our project in the project hierarchy. After saving again we can close out and enter the PCB editor software.

Now we want to make a new drawing and name it "PCB_yourname" and create a new folder inside the project folder we were using earlier, name it accordingly and then select Board(wizard) before clicking ok. On the board wizard window select next until at the general parameters window and choose the units to be millimeters, set the size to A and choose at the center of the drawing button. Select next and select "don't generate artwork films" button, then select next until you get to the spacing constraints window, from here set everything as 0.04, then select the ellipses next to "default via padstack" and type "pad35cir25d" in the space. From here you just need to choose it to be a rectangular board and set the width and height of the board to be 30mm, now you can select finish, and you should see the outline of the PCB.

Now you translate our simulation to PCB by creating a netlist to update our PCB, to start go back to our simulation software and opening our previous simulation. Then we want to go to tools and select create netlist, ensure Create PCB Editor Netlist is checked and click setup. Your configuration file should be C:\Cadence\SPB_22.1\tools\capture\allegro.cfg, if this is the case press OK and then click OK in the "create netlist" window. From here go to the PCB tab and select new layout, then go to setup and select grid and change the grids to have spacing in Non-Etch, Top and Bottom to be 0.15,

do not touch any other settings from here and select OK. Now just go to the place menu and select component manually and then click hide, now place your components as seen in Figure 6 and go to Route -> PCB Router -> Route Automatic and the PCB will automatically be routed.

Now go to Setup -> Cross-section to ensure the number of layers, thickness, material, and other specifications are correct. When done select OK from here we just need to go to Manufacture -> Artwork Control Form, under the film control tab check mark bottom and top, change their undefined line width value to 5 and then click create artwork button to generate the artwork files, ignore all errors and click ok in the artwork control form menu. Now to generate the drill file, to do so just go to manufacture -> NC parameters and checkmark leading zero suppression and enhanced Excellon format. Close the window and go to manufacture -> NC drill and check auto tool select and optimize drill head travel and uncheck all other boxes, click the drill button and close the window.

Now in the PCB Editor go to file -> plot setup and set plot scaling to “fit to page” plot method to “color”, and plot contents to “screen contents” and click OK. Now go to File -> plot and save the design as a PDF, if done correctly it should appear exactly the same as the pdf seen in Figure 6.

Results:

Part A)

- Calculate** the current flowing through each resistor in Figure 1 and fill out the first column of table 1. Show detailed calculation steps in your lab report.
- Build** the circuit in Figure 1 on the breadboard. Measure the current through each resistor using a DMM and complete table 1. Compare the measured data with analytical calculation.

Table 1: Theoretical and Experimental Values for Fig 1. Circuit.

	Theoretical	Experiment
I_{R1}	16.5mA	14.8mA
I_{R2}	13.57mA	12.4mA
I_{R3}	46.0mA	43.02mA
I_{R4}	48.93mA	43.8mA
I_{R5}	2.93mA	2.4mA

Part B)

- Calculate** the Thevenin and Norton circuit model seen from terminals ab in Figure 2. Fill out the second column of Table 2.

- b) **Simulate** the circuit in Figure 2 using PSpice. Show the voltages and currents of each node and branches in your report and fill out the second column of Table 2. Use I_N and V_{th} to derive R_{th} . For what value of R_L the maximum power would be delivered to the load? Use a load R_L and set its value to transfer the maximum power to it. Record the voltage across the load and also the power P_{Lmax} and fill out Table 2.

For the maximum power to be delivered to the load $R_L = 544.9\Omega$.

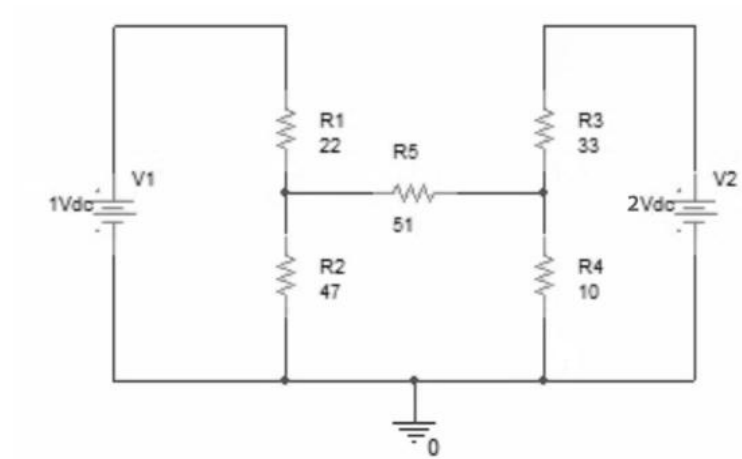
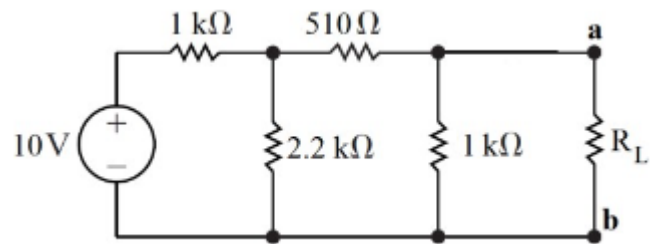
- c) **Build** the circuit in Figure 2 on the breadboard. Measure V_{th} , I_N , and R_{th} using a DMM and complete Table 2. Compare the measured data with simulation and analytical calculation. Use a resistor decade box as load R_L and set its value to transfer the maximum power to it. Measure the voltage across the load and then calculate the power P_{Lmax} and fill out Table 2.
- d) **Design the PCB** of the circuit in Figure 2. Use one layer for routing. Make sure the traces are thick enough.

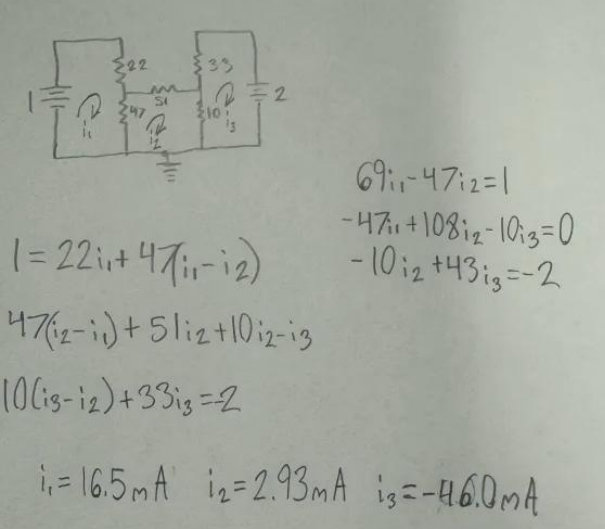
Table 2: Theoretical, Simulated, and Experimental Values for Fig. 2 Circuit.

	Theoretical	Simulation	Experiment
V_{th}	3.1V	2.92V	2.91V
I_N	5.70mA	5.32mA	5.44mA
R_{th}	544.9 Ω	544.9 Ω	534.6 Ω
P_{Lmax}	4.41mW	3.91mW	3.94mW

Conclusion:

In this lab our objective was to independently simulate, create, and calculate the outputs of two different circuits without relying on the guidance of our instructors. Doing so helped us better understand the number of tools we have at our disposal for circuit analysis after having learned methods of analysis over the course of this semester. Doing so without the guidance of our instructor helped us understand not only the new skills at our disposal but also that confidence in your skill plays a significant role in calculations. It is regrettable we had forgotten to photograph our Part B circuit but other than that this lab went by with little to no issue.

Appendix:**Figure 1: Circuit Used for Project A.****Figure 2: Circuit used for Project B.**



Hand-drawn circuit diagram showing a network with three loops and three resistors (22Ω, 47Ω, 10Ω) and two voltage sources (1V, 2V). Currents i_1 , i_2 , and i_3 are indicated through the resistors.

$$\begin{aligned}
 69i_1 - 47i_2 &= 1 \\
 -47i_1 + 108i_2 - 10i_3 &= 0 \\
 -10i_2 + 43i_3 &= -2
 \end{aligned}$$

$$\begin{aligned}
 1 &= 22i_1 + 47(i_1 - i_2) \\
 47(i_2 - i_1) + 51i_2 + 10i_2 - i_3 \\
 10(i_3 - i_2) + 33i_3 &= -2
 \end{aligned}$$

$$i_1 = 16.5 \text{ mA} \quad i_2 = 2.93 \text{ mA} \quad i_3 = -46.0 \text{ mA}$$

Figure 3: Work Done to Find Current Across Each Resistor in Fig. 1.

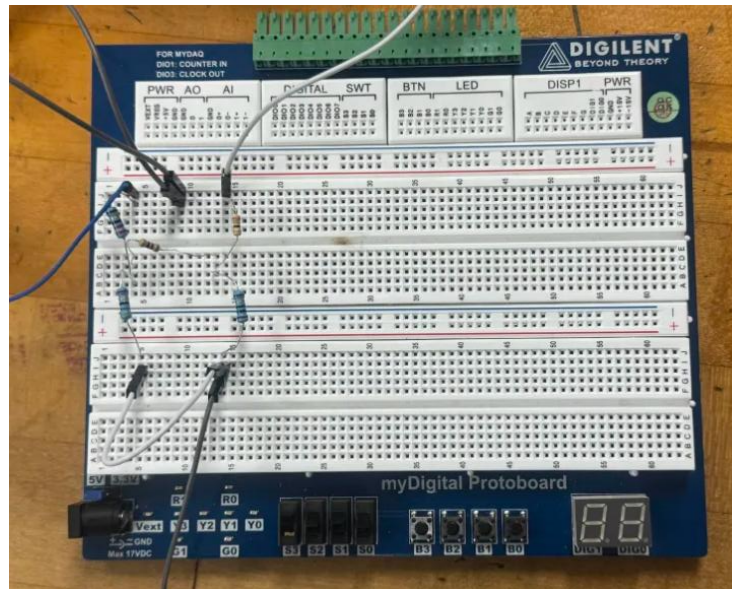


Figure 4: Experimental Circuit from Fig. 1.

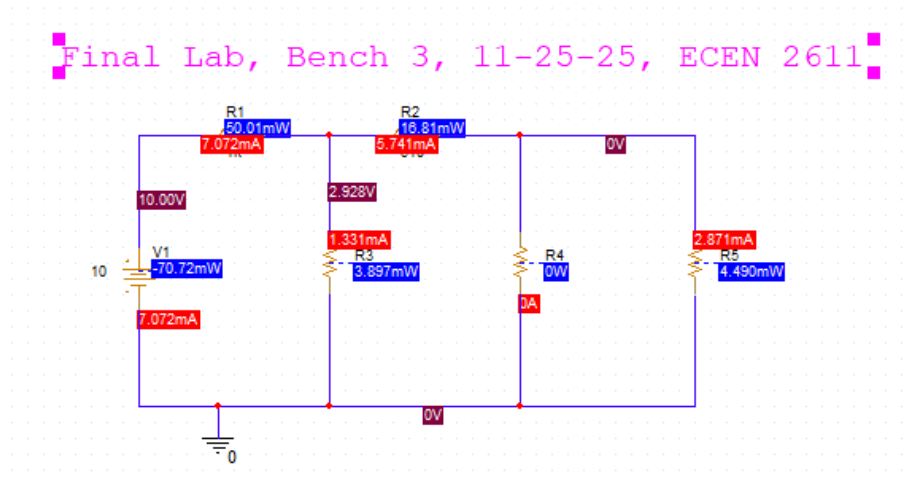


Figure 5: Simulated Circuit from Fig 2.

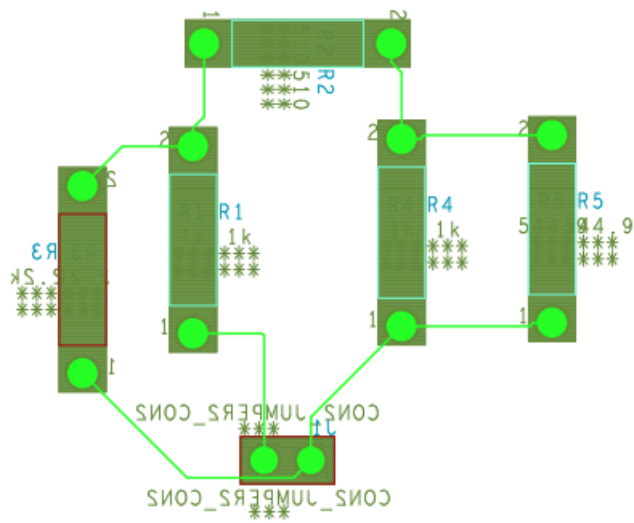


Figure 6: PCB Layout of Circuit Depicted in Figure 2

